

- at the sources of Q_1 and Q_2 ? Express these in terms of the overdrive voltage V_{OV} at which each of Q_1 and Q_2 operates, and V_t .
- (b) For the situation in (a), what current flows in Q_3 ? What overdrive voltage V_{OV3} is Q_3 operating at, in terms of V_C , V_{OV} , and V_t ?
- (c) Now consider the case $v_{G1} = +v_{id}/2$ and $v_{G2} = -v_{id}/2$, where v_{id} is a small signal. Convince yourself that Q_3 now conducts current and operates in the triode region with a small v_{DS} . What resistance r_{DS} does it have, expressed in terms of the overdrive voltage V_{OV3} at which it is operating? This is the resistance R_s . Now if all three transistors have the same W/L , express R_s in terms of V_{OV} , V_{OV3} , and $g_{m1,2}$.
- (d) Find V_{OV3} and hence V_C that result in (i) $R_s = 1/g_{m1,2}$; (ii) $R_s = 0.5/g_{m1,2}$.

***9.22** The circuit of Fig. P9.22 shows an effective way of implementing the resistance R_s needed for the circuit in Fig. P9.20. Here R_s is realized as the series equivalent of two MOSFETs Q_3 and Q_4 that are operated in the triode region, thus, $R_s = r_{DS3} + r_{DS4}$. Assume that Q_1 and Q_2 are matched and operate in saturation at an overdrive voltage V_{OV} that corresponds to a drain bias current of $I/2$. Also, assume that Q_3 and Q_4 are matched.

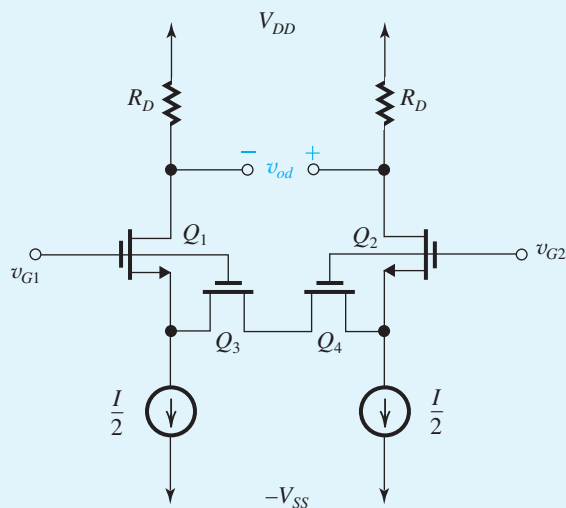


Figure P9.22

- (a) With $v_{G1} = v_{G2} = 0$ V, what dc voltages appear at the sources of Q_1 and Q_2 ? What current flows through Q_3

- and Q_4 ? At what overdrive voltages are Q_3 and Q_4 operating? Find an expression for r_{DS} for each of Q_3 and Q_4 and hence for R_s in terms of $(W/L)_{1,2}$, $(W/L)_{3,4}$, and $g_{m1,2}$.
- (b) Now with $v_{G1} = v_{id}/2$ and $v_{G2} = -v_{id}/2$, where v_{id} is a small signal, find an expression of the voltage gain $A_d \equiv v_{od}/v_{id}$ in terms of $g_{m1,2}$, R_D , $(W/L)_{1,2}$, and $(W/L)_{3,4}$.

D *9.23 Figure P9.23 shows a circuit for a differential amplifier with an active load. Here Q_1 and Q_2 form the differential pair, while the current source transistors Q_4 and Q_5 form the active loads for Q_1 and Q_2 , respectively. The dc bias circuit that establishes an appropriate dc voltage at the drains of Q_1 and Q_2 is not shown. It is required to design the circuit to meet the following specifications:

- (a) Differential gain $A_d = 50$ V/V.
 (b) $I_{REF} = I = 200$ μ A.
 (c) The dc voltage at the gates of Q_6 and Q_3 is $+0.8$ V.
 (d) The dc voltage at the gates of Q_7 , Q_4 , and Q_5 is -0.8 V.

The technology available is specified as follows: $\mu_n C_{ox} = 2.5$ μ A/V²; $V_{tn} = |V_{tp}| = 0.5$ V, $V_{An} = |V_{Ap}| = 10$ V. Specify the required value of R and the W/L ratios for all transistors. Also specify I_D and $|V_{GS}|$ at which each transistor is operating. For dc bias calculations you may neglect channel-length modulation.

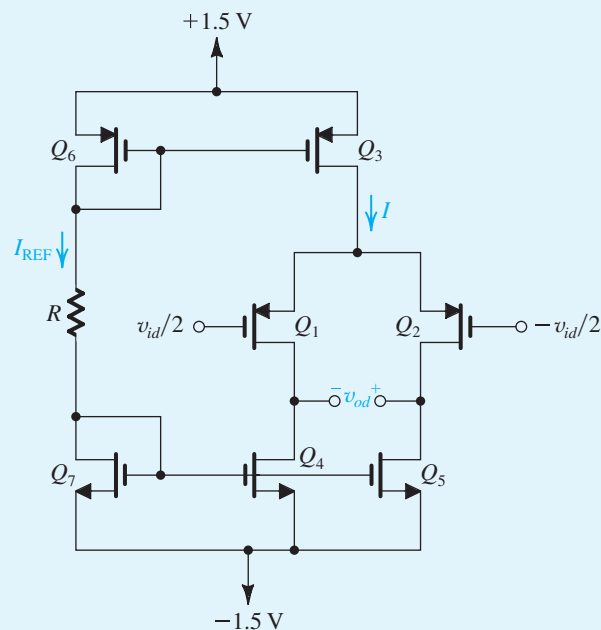


Figure P9.23

the input offset voltage is approximately $V_T[(1/\beta_1) - (1/\beta_2)]$. Evaluate V_{OS} for $\beta_1 = 50$ and $\beta_2 = 100$.

9.78 Two possible differential amplifier designs are considered, one using BJTs and the other MOSFETs. In both cases, the collector (drain) resistors are maintained within $\pm 2\%$ of nominal value. The MOSFETs are operated at $V_{OV} = 200$ mV. What input offset voltage results in each case? What does the MOS V_{OS} become if the devices are increased in width by a factor of 4 while the bias current is kept constant?

***9.79** A differential amplifier uses two transistors having V_A values of 100 V and 200 V. If everything else is matched, find the resulting input offset voltage. Assume that the two transistors are intended to be biased at a V_{CE} of about 10 V.

***9.80** A differential amplifier is fed in a balanced or push-pull manner, and the source resistance in series with each base is R_s . Show that a mismatch ΔR_s between the values of the two source resistances gives rise to an input offset voltage of approximately $(I/2\beta)\Delta R_s / [1 + (g_m R_s)/\beta]$.

9.81 One approach to “offset correction” involves the adjustment of the values of R_{C1} and R_{C2} so as to reduce the differential output voltage to zero when both input terminals are grounded. This offset-nulling process can be accomplished by utilizing a potentiometer in the collector circuit, as shown in Fig. P9.81. We wish to find the

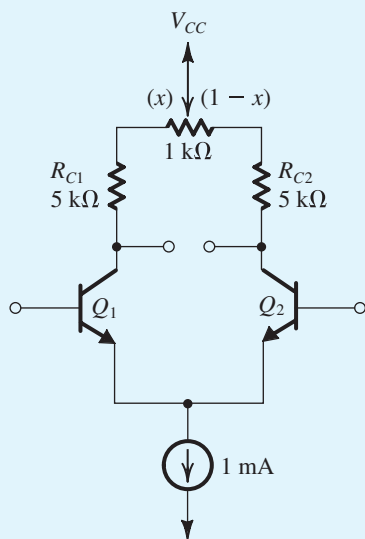


Figure P9.81

potentiometer setting, represented by the fraction x of its value connected in series with R_{C1} , that is required for nulling the output offset voltage that results from:

- (a) R_{C1} being 4% higher than nominal and R_{C2} 4% lower than nominal
- (b) Q_1 having an area 5% larger than nominal, while Q_2 has area 5% smaller than nominal.

9.82 A differential amplifier for which the total emitter bias current is 400 μ A uses transistors for which β is specified to lie between 80 and 200. What is the largest possible input bias current? The smallest possible input bias current? The largest possible input offset current?

****9.83** In a particular BJT differential amplifier, a production error results in one of the transistors having an emitter-base junction area twice that of the other. With both inputs grounded, find the current in each of the two transistors and hence the dc offset voltage at the output, assuming that the collector resistances are equal. Use small-signal analysis to find the input voltage that would restore current balance to the differential pair. Repeat using large-signal analysis and compare results.

D 9.84 A large fraction of mass-produced differential-amplifier modules employing 20-k Ω collector resistors is found to have an input offset voltage ranging from +2 mV to -2 mV. By what amount must one collector resistor be adjusted to reduce the input offset to zero? If an adjustment mechanism is devised that raises one collector resistance while correspondingly lowering the other, what resistance change is needed? If a potentiometer connected as shown in Fig. P9.81 is used, what value of potentiometer resistance (specified to 1 significant digit) is needed? Assume that the offset is entirely due to the finite tolerance of R_C .

Section 9.5: The Differential Amplifier with a Current-Mirror Load

9.85 The differential amplifier of Fig. 9.32(a) is measured and found to have a short-circuit transconductance of 2 mA/V. A differential input signal is applied and the output voltage is measured with a load resistance R_L connected. It is found that when R_L is reduced from ∞ to 20 k Ω , the magnitude of the output signal is reduced by half. What do you estimate R_o and A_d (with R_L disconnected) to be?

9.86 A current-mirror-loaded NMOS differential amplifier is fabricated in a technology for which $|V'_A| = 5$ V/ μ m. All the

transistors have $L = 0.5 \mu\text{m}$. If the differential-pair transistors are operated at $V_{OV} = 0.25 \text{ V}$, what open-circuit differential gain is realized?

9.87 The differential amplifier of Fig. 9.32(a) is biased with $I = 200 \mu\text{A}$. All transistors have $L = 0.5 \mu\text{m}$, and Q_1 and Q_2 have $W/L = 50$. The circuit is fabricated in a process for which $\mu_n C_{ox} = 200 \mu\text{A/V}^2$ and $|V_A| = 5 \text{ V}/\mu\text{m}$. Find $g_{m1,2}$, r_{o2} , r_{o4} , and A_d .

D 9.88 In a current-mirror-loaded differential amplifier of the form shown in Fig. 9.32(a), all transistors are characterized by $k'W/L = 4 \text{ mA/V}^2$, and $|V_A| = 5 \text{ V}$. Find the bias current I for which the gain $v_o/v_{id} = 20 \text{ V/V}$.

D 9.89 Consider a current-mirror-loaded differential amplifier such as that shown in Fig. 9.32(a) with the bias current source implemented with the modified Wilson mirror of Fig. P9.89 with $I = 200 \mu\text{A}$. The transistors have $|V_t| = 0.5 \text{ V}$ and $k'W/L = 5 \text{ mA/V}^2$. What is the lowest value of the total power supply ($V_{DD} + V_{SS}$) that allows each transistor to operate with $|V_{DS}| \geq |V_{GS}|$?

***9.90** (a) Sketch the circuit of a current-mirror-loaded MOS differential amplifier in which the input transistors are cascoded and a cascode current mirror is used for the load.

(b) Show that if all transistors are operated at an overdrive voltage V_{OV} and have equal Early voltages $|V_A|$, the gain is given by

$$A_d = 2(V_A/V_{OV})^2$$

Evaluate the gain for $V_{OV} = 0.20 \text{ V}$ and $V_A = 10 \text{ V}$.

9.91 Figure P9.91 shows the current-mirror-loaded MOS differential amplifier prepared for small-signal analysis. We have “pulled out” r_o of each transistor; thus, the current in the drain of each transistor will be $g_m v_{gs}$. To help the reader, we have already indicated approximate values for some of the node voltages. For instance, the output voltage $v_o = \frac{1}{2}(g_m r_o) v_{id}$, which we have derived in the text. The voltage at the common sources has been found to be approximately $+v_{id}/4$, which is very far from the virtual ground one might assume. Also, the voltage at the gate of the mirror is approximately $-v_{id}/4$, confirming our contention that the voltage there is vastly different from the output voltage, hence the lack of balance in the circuit and the unavailability of a differential half-circuit. Find the currents labeled i_1 to i_{13} in terms of $(g_m v_{id})$. Determine their values in the sequence of their numbering and assume $g_m r_o \gg 1$. Note that all transistors are assumed to be operating at the same $|V_{OV}|$. Write the current values on the circuit diagram and reflect on the results.

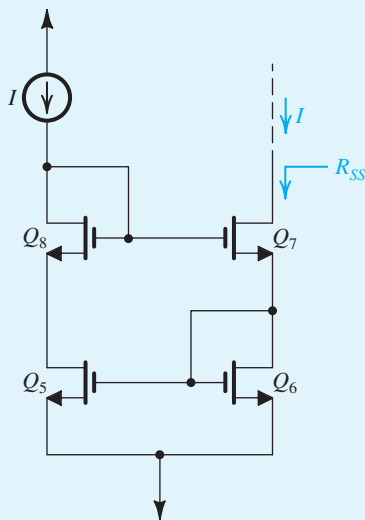


Figure P9.89

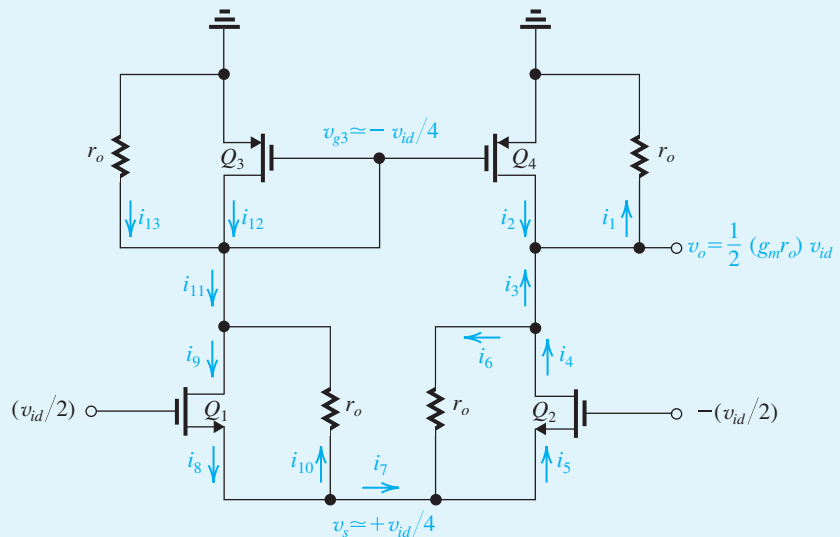


Figure P9.91

9.92 A current-mirror-loaded NMOS differential amplifier operates with a bias current I of 200 μA . The NMOS transistors are operated at $V_{OV} = 0.2\text{ V}$ and the PMOS devices at $|V_{OV}| = 0.3\text{ V}$. The Early voltages are 20 V for the NMOS and 12 V for the PMOS transistors. Find G_m , R_o , and A_d . For what value of load resistance is the gain reduced by a factor of 2?

9.93 This problem investigates the effect of transistor mismatches on the input offset voltage of the current-mirror-loaded MOS differential amplifier of Fig. 9.32(a). For this purpose, ground both input terminals and short-circuit the output node to ground.

(a) If the amplifying transistors Q_1 and Q_2 exhibit a W/L mismatch of $\Delta(W/L)_A$, find the resulting short-circuit output current and hence show that the corresponding V_{OS} is given by

$$V_{OS1} = (V_{OV}/2) \frac{\Delta(W/L)_A}{(W/L)_A}$$

where V_{OV} is the overdrive voltage at which Q_1 and Q_2 are operating.

(b) Repeat for a mismatch $\Delta(W/L)_M$ in the W/L ratios of the mirror transistor Q_3 and Q_4 to show that the corresponding V_{OS} is given by

$$V_{OS2} = (V_{OV}/2) \frac{\Delta(W/L)_M}{(W/L)_M}$$

where V_{OV} is the overdrive voltage at which Q_1 and Q_2 are operating.

(c) For a circuit in which all transistors are operated at $|V_{OV}| = 0.2\text{ V}$ and all W/L ratios are accurate to within $\pm 1\%$ of nominal, find the worst-case total offset voltage V_{OS} .

9.94 The differential amplifier in Fig. 9.36(a) is operated with $I = 500\text{ }\mu\text{A}$, with devices for which $V_A = 10\text{ V}$ and $\beta = 100$. What differential input resistance, output resistance, short-circuit transconductance, and open-circuit voltage gain would you expect? What will the voltage gain be if the input resistance of the subsequent stage is equal to R_{id} of this stage?

9.95 A bipolar differential amplifier having a simple pnp current-mirror load is found to have an input offset voltage of 2 mV. If the offset is attributable entirely to the finite β of the pnp transistors, what must β_p be?

9.96 For the current-mirror-loaded bipolar differential pair, replacing the simple current-mirror load by the base-current-compensated mirror of Fig. 8.11, find the expected systematic input offset voltage. Evaluate V_{OS} for $\beta_p = 50$.

9.97 For the current-mirror-loaded bipolar differential pair, replacing the simple current-mirror load by the Wilson mirror of Fig. 8.40(a), find the expected systematic input offset voltage. Evaluate V_{OS} for $\beta_p = 50$.

9.98 Figure P9.98 shows a differential cascode amplifier with an active load formed by a Wilson current mirror. Utilizing the expressions derived in Chapter 8 for the output resistance of a bipolar cascode and the output resistance of the Wilson mirror, and assuming all transistors to be identical, show that the differential voltage gain A_d is given approximately by

$$A_d = \frac{1}{3} \beta g_m r_o$$

Evaluate A_d for the case of $\beta = 100$ and $V_A = 20\text{ V}$.

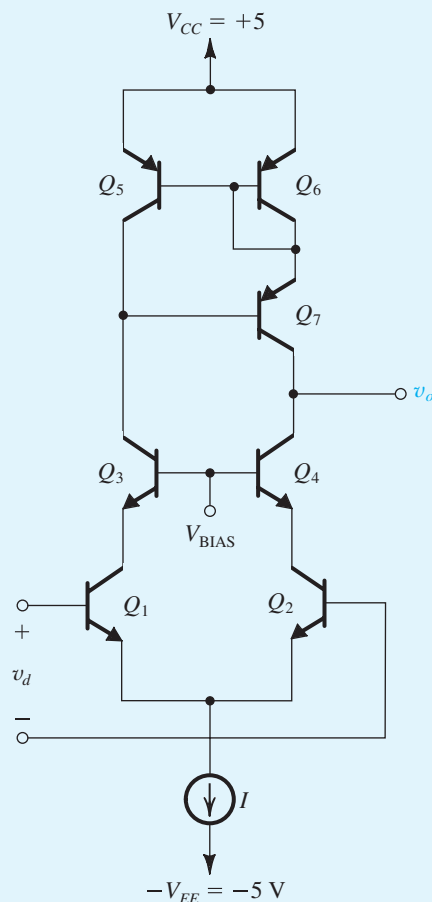


Figure P9.98

D 9.99 Consider the bias design of the Wilson-loaded cascode differential amplifier shown in Fig. P9.98.

- What is the largest signal voltage possible at the output without Q_7 saturating? Assume that the CB junction conducts when the voltage across it exceeds 0.4 V.
- What should the dc bias voltage established at the output (by an arrangement not shown) be in order to allow for positive output signal swing of 1.5 V?
- What should the value of V_{BIAS} be in order to allow for a negative output signal swing of 1.5 V?
- What is the upper limit on the input common-mode voltage v_{CM} ?

****9.100** Figure P9.100 shows a modified cascode differential amplifier. Here Q_3 and Q_4 are the cascode transistors. However, the manner in which Q_3 is connected with its base current feeding the current mirror Q_7 – Q_8 results in very interesting input properties. Note that for simplicity the circuit is shown with the base of Q_2 grounded. Assume that all transistors have equal β 's.

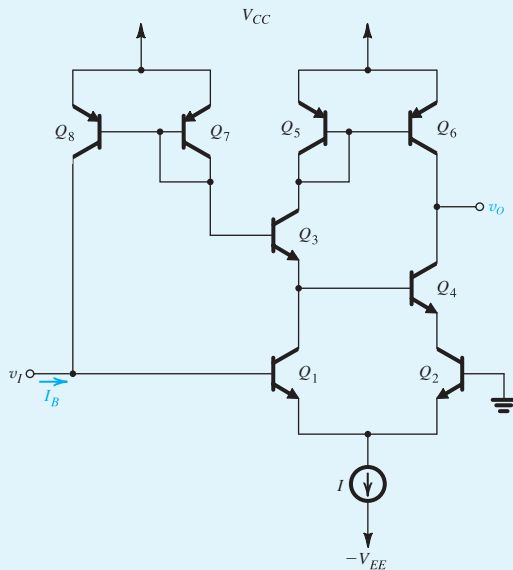


Figure P9.100

- With $v_I = 0$ V dc, find the input bias current I_B assuming all transistors have equal value of β . Compare with the case without the Q_7 – Q_8 connection.

- With $v_I = 0$ V (dc) + v_{id} , find the input signal current i_i and hence the input differential resistance R_{id} . Compare with the case without the Q_7 – Q_8 connection. By what factor does R_{id} increase?

9.101 For the folded-cascode differential amplifier of Fig. 9.38, find the value of V_{BIAS} that results in the largest possible positive output swing, while keeping Q_3 , Q_4 , and the *pnp* transistors that realize the current sources out of saturation. Assume $V_{\text{CC}} = V_{\text{EE}} = 5$ V. If the dc level at the output is 0 V, find the maximum allowable output signal swing. For $I = 0.5$ mA, $\beta_P = 50$, $\beta_N = 100$, and $V_A = 100$ V find G_m , R_{o4} , R_{o5} , R_o , and A_d .

9.102 For the BiCMOS differential amplifier in Fig. P9.102 let $V_{\text{DD}} = V_{\text{SS}} = 3$ V, $I = 0.2$ mA, $k'_p W/L = 6.4$ mA/V²; $|V_A|$ for *p*-channel MOSFETs is 10 V, $|V_A|$ for *n*pn transistors is 30 V. Find G_m , R_o , and A_d .

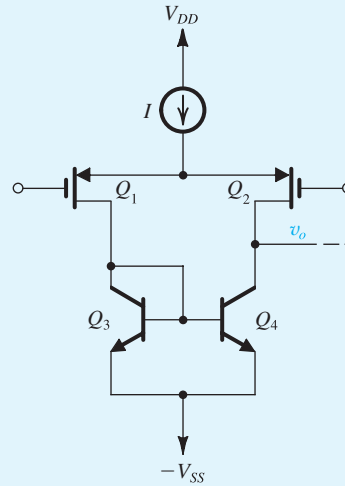


Figure P9.102

SIM D 9.103 It is required to design the current-mirror-loaded differential MOS amplifier of Fig. 9.32 to obtain a differential gain of 50 V/V. The technology available provides $\mu_n C_{ox} = 4\mu_p C_{ox} = 400$ $\mu\text{A/V}^2$, $|V_t| = 0.5$ V, and $|V'_A| = 20$ V/ μm and operates from ± 1 V supplies. Use a bias current $I = 200$ μA and operate all devices at $|V_{\text{OV}}| = 0.2$ V.

- Find the W/L ratios of the four transistors.
- Specify the channel length required of all transistors.
- If $V_{\text{ICM}} = 0$, what is the allowable range of v_O ?

- (d) If I is delivered by a simple NMOS current source operated at the same V_{OV} and having the same channel length as the other four transistors, determine the CMRR obtained.

9.104 Consider the current-mirror-loaded MOS differential amplifier of Fig. 9.32(a) in two cases:

- (a) Current source I is implemented with a simple current mirror.
 (b) Current source I is implemented with the modified Wilson current mirror shown in Fig. P9.104.

Recalling that for the simple mirror $R_{SS} = r_{o1} \parallel r_{o5}$ and for the Wilson mirror $R_{SS} \simeq g_{m7} r_{o7} r_{o5}$, and assuming that all transistors have the same $|V_A|$ and $k'W/L$, show that for case (a)

$$\text{CMRR} = 2 \left(\frac{V_A}{V_{OV}} \right)^2$$

and for case (b)

$$\text{CMRR} = 2\sqrt{2} \left(\frac{V_A}{V_{OV}} \right)^3$$

where V_{OV} is the overdrive voltage that corresponds to a drain current of $I/2$. For $k'W/L = 4 \text{ mA/V}^2$, $I = 160 \mu\text{A}$, and $|V_A| = 5 \text{ V}$, find CMRR for both cases.

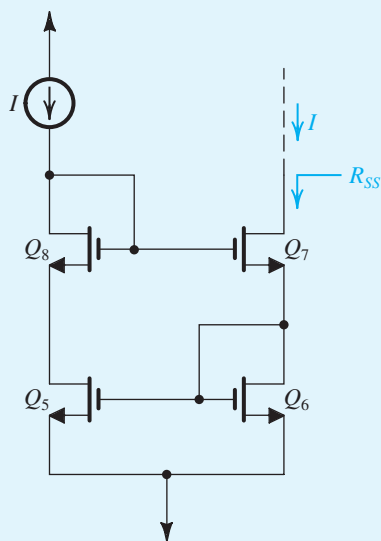


Figure P9.104

9.105 The MOS differential amplifier of Fig. 9.32(a) is biased with a simple current mirror delivering $I = 200 \mu\text{A}$. All transistors are operated at $V_{OV} = 0.2 \text{ V}$ and have $V_A = 5 \text{ V}$. Find G_m , R_o , A_d , R_{SS} , G_{mcm} , R_{im} , A_m , R_{om} , R_{o2} , A_{cm} , and CMRR.

9.106 A current-mirror-loaded MOS differential amplifier is found to have a differential voltage gain A_d of 30 V/V . Its bias current source has an output resistance $R_{SS} = 45 \text{ k}\Omega$. The current mirror utilized has a current gain A_m of 0.98 A/A and an output resistance R_{om} of $45 \text{ k}\Omega$. If the common-mode output resistances of the amplifier, R_{o1} and R_{o2} , are very large, find A_{cm} and CMRR.

9.107 A current-mirror-loaded MOS differential amplifier is found to have a differential voltage gain A_d of 50 V/V and a CMRR of 60 dB . If the output resistance of the bias current source is $20 \text{ k}\Omega$ and the output resistance of the current-mirror load is $20 \text{ k}\Omega$, what is the expected magnitude of the deviation from unity of the current gain of the load mirror?

D *9.108 Design the circuit of Fig. 9.36(a) using a basic current mirror to implement the current source I . It is required that the short-circuit transconductance be 5 mA/V . Use $\pm 5\text{-V}$ power supplies and BJTs that have $\beta = 100$ and $V_A = 100 \text{ V}$. Give the complete circuit with component values and specify the differential input resistance R_{id} , the output resistance R_o , the open-circuit voltage gain A_d , the input bias current, the input common-mode range, the common-mode gain, and the CMRR.

D *9.109 Repeat the design of the amplifier specified in Problem 9.108 utilizing a Widlar current source (Fig. 8.42) to supply the bias current. Assume that the largest resistance available is $2 \text{ k}\Omega$.

9.110 A bipolar differential amplifier such as that shown in Fig. 9.36(a) has $I = 0.4 \text{ mA}$, $V_A = 40 \text{ V}$, and $\beta = 150$. Find G_m , R_o , A_d , and R_{id} . If the bias current source is implemented with a simple $n\text{pn}$ current mirror, find R_{EE} , A_{cm} , and CMRR. If the amplifier is fed differentially with a source having a total of $30 \text{ k}\Omega$ resistance (i.e., $15 \text{ k}\Omega$ in series with the base lead of each of Q_1 and Q_2), find the overall differential voltage gain.

9.111 For the current-mirror-loaded differential pair in Fig. P9.111, find:

- differential input resistance, R_{id}
- A_d
- CMRR

Assume $\beta = 100$, $|V_{BE}| = 0.7$ V, and $|V_A| = 60$ V.

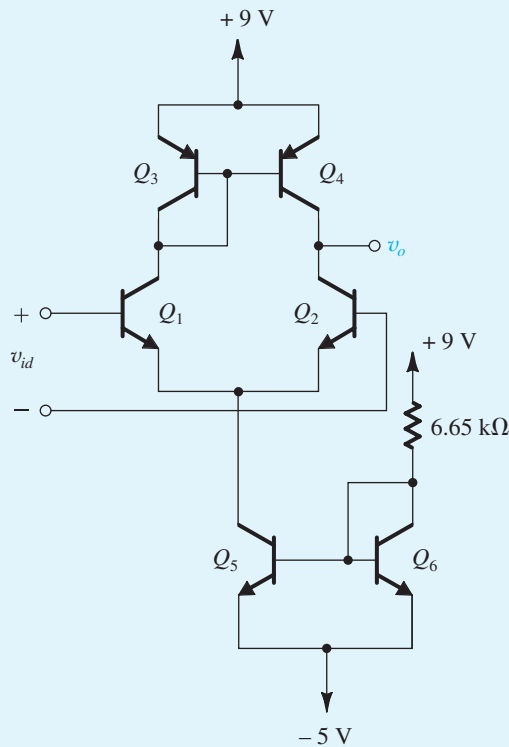


Figure P9.111

9.112 For the current-mirror-loaded differential amplifier in Fig. P9.112, find:

- differential input resistance, R_{id}
- A_d
- CMRR

Assume $\beta = 100$, $|V_{BE}| = 0.7$ V, $|V_A| = 60$ V, $V_t = 0.7$ V, and $k'(W/L) = 2$ mA/V².

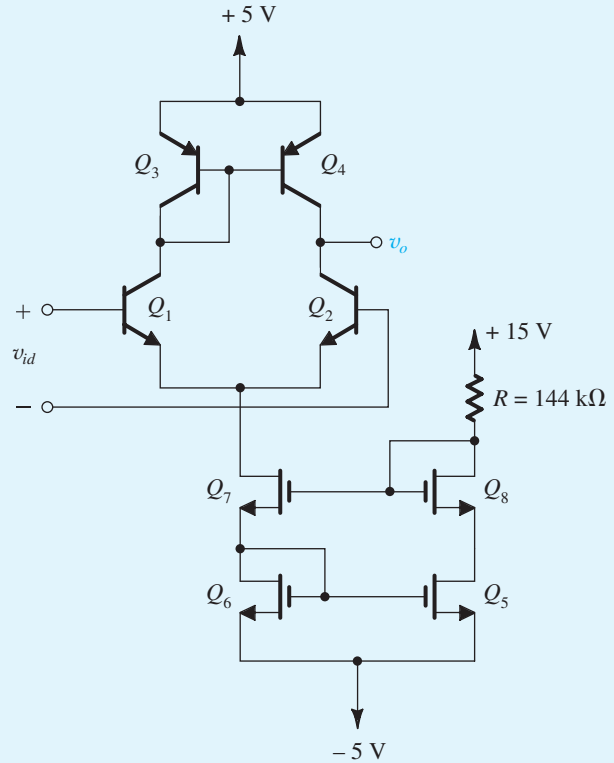


Figure P9.112

Section 9.6: Multistage Amplifiers

9.113 Consider the circuit in Fig. 9.40 with the device geometries (in μm) shown in Table P9.113. Let $I_{\text{REF}} = 225 \mu\text{A}$, $|V_t| = 0.75$ V for all devices, $\mu_n C_{ox} = 180 \mu\text{A/V}^2$, $\mu_p C_{ox} = 60 \mu\text{A/V}^2$, $|V_A| = 9$ V for all devices, $V_{DD} = V_{SS} = 1.5$ V. Determine the width of Q_6 , W , that will ensure that the op amp will not have a systematic offset voltage. Then, for all devices evaluate I_D , $|V_{ov}|$, $|V_{gs}|$, g_m , and r_o . Provide your results in a table similar to Table 9.1. Also find A_1 , A_2 , the open-loop voltage gain, the input